



Program and Branch Name:

[B.Tech CSE-A]

PBL Progress Report – I

Field	Details
Course Name	Data Structure and Algorithms - II
Course Code	CCSE0301
Faculty Name	Mr. Shamshad Ali
Assignment Type	Individual Assignment
Student(s) Name	Afjal Ansari
Roll No. (II Year onwards)	2501330100042
Team/Group No. (Group/Capstone only)	N/A
Problem Statement / Project Title	Intelligent Election Management System
SDG(s)	SDG 16 - Peace, Justice & Strong Institutions
Reporting Month	Month 1
Date of Submission	12/09/2026

Student Declaration

I certify that this assignment is my own work, with all sources appropriately acknowledged. I understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarised or inappropriately AI-generated.

Student Name & Signature: Afjal Ansari

Date: 09/09/2026

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Review 1: Problem Understanding & Initial Progress

1. Problem Statement	Intelligent Election Management System Traditional elections suffer from manual inefficiencies, logistical bottlenecks, and fraud vulnerabilities. These flaws cause long wait times, delayed results, and reduced public trust. Administrators lack a centralized platform to efficiently allocate resources and securely authenticate voters.
2. Problem Context & Background	Traditional voting systems are susceptible to manual errors, inefficiencies in vote counting, and transparency issues. An intelligent system is required to digitize, secure, and streamline the election process for better accessibility and trust.
3. Problem Identification / Understanding	There is a need for a secure authentication mechanism and real-time result tracking in conventional electoral methods. The proposed system will resolve these issues using a robust web-based architecture (e.g., FastAPI, React, PostgreSQL) to ensure data integrity.
4. Objectives	- To develop a secure, user-friendly voting portal. - To implement real-time tracking of voter turnout and results. - To ensure strict voter authentication and transparency.
5. Target Users / Stakeholders	Voters, Election Administrators/Commission, Candidates, and System Auditors.
6. Research / Literature Review Completed	1. A New Era of Elections: Leveraging Blockchain for Fair and Transparent Voting • Link: arXiv:2502.16127 2. Artificial Intelligence for Electoral Management • Link: International IDEA Publication 3. Electoral Security and Artificial Intelligence • Link: Electoral Security and Artificial Intelligence
7. Unit/Module Concepts Identified for Application	Graphs, Queue, Priority Queues (Heaps), Arrays, Linked Lists, Breadth First Search (BFS), Depth First Search (DFS).
8. Work Completed During Month 1	- Finalized project scope and problem statement. - Completed initial requirement gathering and stakeholder analysis.

9. Evidence / Supporting Material	Topic Approval Form, literature review notes, algorithm study materials, project planning documents, and Topic Tree. GitHub Repository Link
10. Challenges Faced	Understanding the complex security constraints required to guarantee voter anonymity while accurately tallying results without duplications.
11. Individual	Conducted requirement analysis, literature review, and drafted the initial architecture and database schema for the system. (Individual Contribution = 100%)
12. Overall Progress (%)	25%
13. Plan for Next Review	Develop the frontend user dashboards (Admin & Voter panels) and initialize the backend relational database schemas.

Faculty Evaluation: To be completed by Faculty in Moodle